



GENED DX2

(VYG) / PB (SPLE)



with COACH TOTO





LET & CSE
#1 ONLINE REVIEW AUTHORITY



Gurong Pinoy



PRAYER



Dear Lord, I come to you to ask for your guidance and direction in this study session. I ask that the holy spirit fill me with strength, creativity and understanding to get through my studies without difficulty or sin. Help me hold my focus and attention. Help me to retain all that I learned. Please make my mind sharp and keep distractions at bay. If any part of my studying is weak or lacking in some way, let it be revealed so that I may correct it. Thank you for this opportunity to learn. Amen.

**MATH ka lang...
Magiging**

LPT

AKO!!



1. In Roman numerals, what letter in the alphabet represents 1,000?

a. M

b. L

c. D

d. C

ROMAN NUMERALS CHART

1	I
2	II
3	III
4	IV
5	V
6	VI
7	VII
8	VIII
9	IX
10	X

11	XI
20	XX
30	XXX
40	XL
50	L
60	LX
70	LXX
80	LXXX
90	XC
100	C

200	CC
300	CCC
400	CD
500	D
600	DC
700	DCC
800	DCCC
900	CM
1000	M
1001	MI

I	V	X	L	C	D	M
1	5	10	50	100	500	1000

2. The value of the expression $3xy + 3y + 7$ when $x = 2$ and $y = 3$ is:

a. 36

b. 42

 c. 34

d. 40

$$3xy + 3y + 7$$

$$= 3(2)(3) + 3(3) + 7$$

$$= 18 + 9 + 7$$

$$= 34$$

3. A box holds 222 apples. How many apples are there in 20 boxes?

- a. 4400
- ☒ b. 4440
- c. 4420
- d. 4460

20 boxes, 222 apples per box

$$= 20 \text{ boxes} \times 222 \frac{\text{apples}}{\text{box}}$$

$$= 4440 \text{ apples}$$



4. What is the amount due on P 12,000.00 after adding the annual interest of 8% for 3 years?

- ✓ a. P14, 880.00
- b. P16, 800.00
- c. P15, 880.00
- d. P15, 800.00

$$FA = 12,000 (1 + (0.08 \times 3))$$

$$FA = 12,000 \{1 + 0.24\}$$

$$FA = 12,000 (1.24)$$

$$FA = 14,880$$

Simple interest formula

$$A = P (1 + rt)$$

A = Final amount

r = Annual interest rate

P = Initial principal balance

t = Time (in years)

INSIDER

5. If the student has an average of 76% on his first two tests and has an average of 85% on the next 4 tests, what is the final average?

- a. 82.0%
- b. 81.3%
- c. 82.59%
- d. 82.5%

$Ave_1 = 76\%$ - two tests

$Ave_2 = 85\%$ - four tests

$$Ave_F = \frac{2(76) + 4(85)}{2 + 4}$$

$$Ave_F = \frac{492}{6}$$

$$Ave_F = 82\%$$

Average Formula



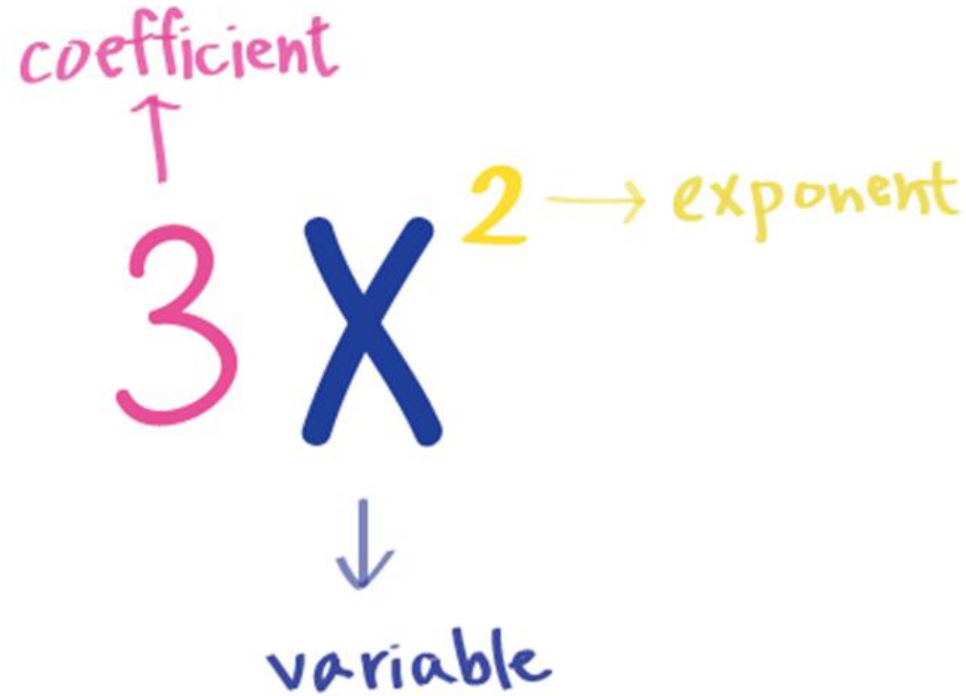
$$\begin{aligned} \text{Average} &= \frac{\text{Sum of all observations}}{\text{Total number of observations}} \\ &= \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \end{aligned}$$

$a_1, a_2, \dots, a_n \rightarrow$ Observations

$n \rightarrow$ Number of observations

6. What is the coefficient in the expression $18x^3y^3$?

- ☒ a. 18
- b. X
- c. 3
- d. 1



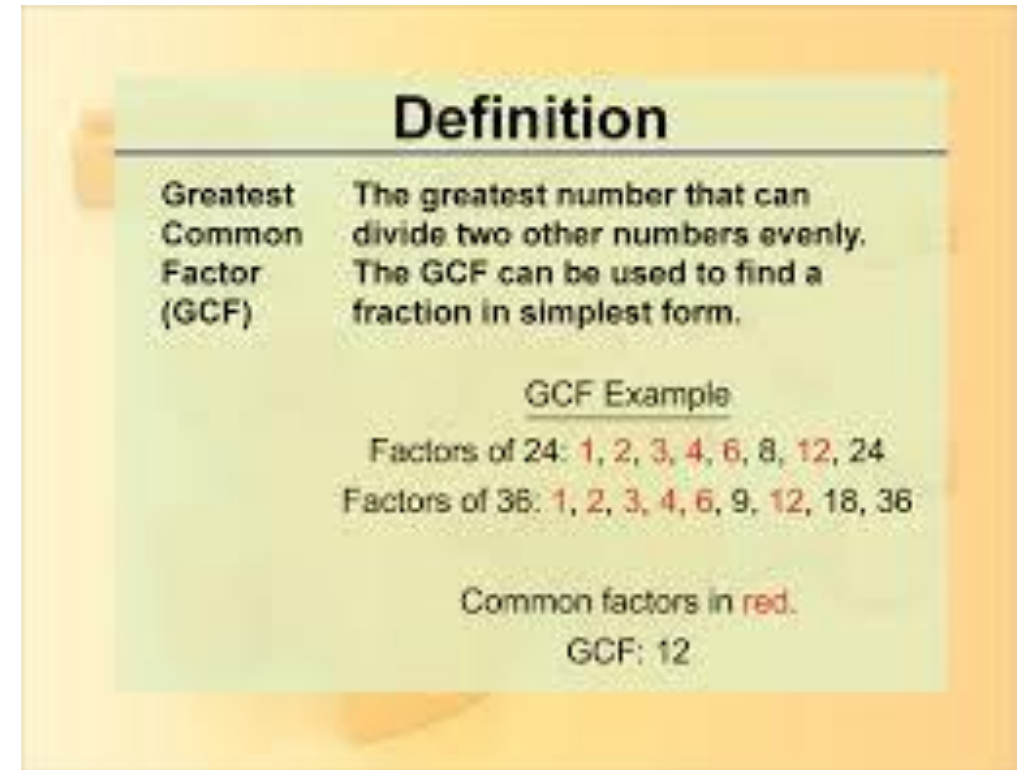
7. Simplify: $(2y^2+4y-5xy) + (8xy+2y^2+3y)$

-  a. $4y^2+7y+3xy$
- b. $-4y^2+2y-3xy$
- c. $4y^2-7y-3xy$
- d. $4y^2-7y+3xy$

$$\begin{array}{r} 2y^2 + 4y - 5xy \\ + \quad 2y^2 + 3y + 8xy \\ \hline 4y^2 + 7y + 3xy \end{array}$$

8. The greatest common factor of 52 and 104 is ____

- a. 9
- ☒ b. 13
- c. 11
- d. 7



Definition

Greatest Common Factor (GCF)	The greatest number that can divide two other numbers evenly. The GCF can be used to find a fraction in simplest form.
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GCF Example

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24

Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36

Common factors in red.

GCF: 12

(GCF is 52 but not in the choices)

9. Find the amount due at the end of 2 ½ years if P 8,000.00 is invested at 12% simple interest.

- ✓ a. P10,400.00
- b. P12, 400.00
- c. P14,800.00
- d. P13,800.00

$$FA = 8,000 [1 + (0.12 \times 2.5)]$$

$$FA = 8,000 [1 + (0.3)]$$

$$FA = 8,000 [1.3]$$

$$FA = 10,400$$

Simple interest formula

$$A = P(1 + rt)$$

A = Final amount

r = Annual interest rate

P = Initial principal balance

t = Time (in years)

10. Simplify: $(2x-3y) + (y-4z) - (z-3x)$

a. $5x+2y-5z$

b. $-5x+2y+5z$

c. $5x+2y+5z$

 d. $5x-2y-5z$

$$(2x-3y) + (y-4z) - (z-3x)$$

$$2x - 3y + y - 4z - z + 3x$$

$$5x - 2y - 5z$$

11. How much was borrowed if the interest at 8% is P6,000 after 3 months is 76,000.00?

- a. P25,000.00
- ✓ b. P300,000.00
- c. P150,000.00
- d. P200,000.00

$$I = 6,000$$

$$r = 8\% \text{ or } 0.08$$

$$t = 3/12 \text{ or } 0.25$$

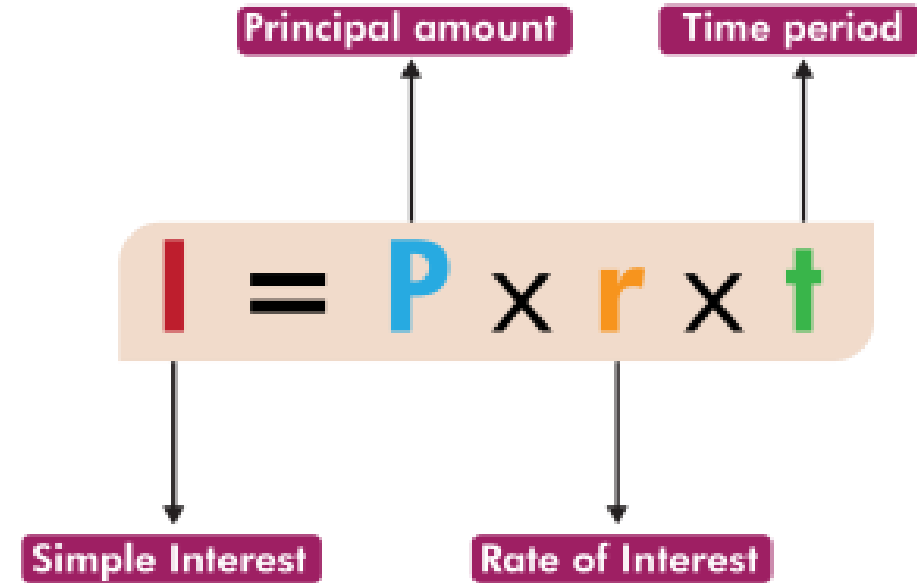
$$P = ?$$

$$6,000 = P \times 0.08 \times 0.25$$

$$6,000 = P \times 0.02$$

$$\frac{6,000}{0.02} = \frac{0.02P}{0.02}$$

$$300,000 = P$$



12. What is the amount due on P10,000.00 after adding the annual interest of 8% for 3 years?

- a. P 13,400.00
 - b. P 11,800.00
 - ✓ c. P12,400.00
 - d. P 15,800.00
- $P = 10,000$
 $r = 8\% \text{ or } 0.08$
 $t = 3$

Simple interest formula

$$A = P (1 + rt)$$

A = Final amount

r = Annual interest rate

P = Initial principal balance

t = Time (in years)

INSIDER

$$FA = 10,000 [1 + (0.08 \times 3)]$$

$$FA = 10,000 [1 + 0.24]$$

$$FA = 10,000 [1.24]$$

$$FA = 12,400$$

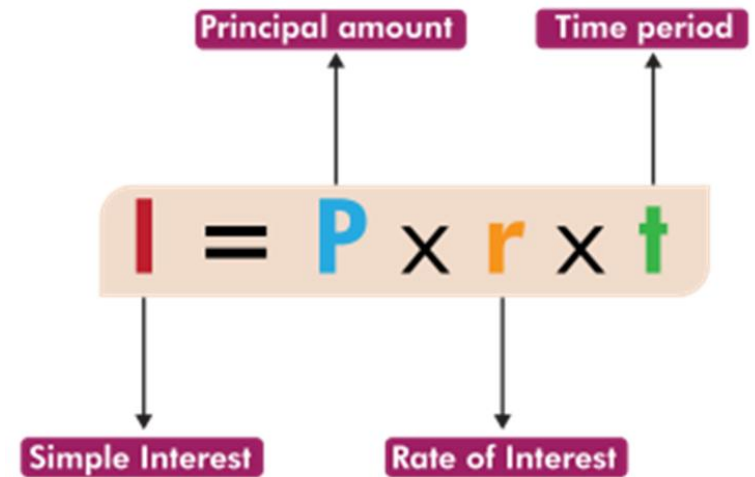
13. The simple interest on P20,000.00 for 3 years at 9% annually is ____

- a. P 6,200.00
- ☒ b. P 5,400.00
- c. P 6,800.00
- d. P 4,800.00

P = 20,000
R = 9% or 0.09
T = 3 years
I = ?

$$I = 20,000 \times 0.09 \times 3$$

$$I = 5,400$$



14. Vivian obtained the following results from her Mathematics examinations: 90, 83, 82, 85. What score must she get on the next examination so that her average score is 86?

A. 84

B. 89

☒ C. 90

D. 93

$$86 = \frac{90+83+82+85+X}{5}$$

$$5 \left[86 = \frac{90+83+82+85+X}{5} \right] 5$$

$$430 = 340 + X$$

$$430 - 340 = 340 - 340 + X$$

$$90 = X$$

Average Formula



$$\text{Average} = \frac{\text{Sum of all observations}}{\text{Total number of observations}}$$

$$= \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

$a_1, a_2, \dots, a_n \rightarrow$ Observations

$n \rightarrow$ Number of observations

15. How long will it take A and B together, to finish a job which can be done by A alone in 6 days and B alone in 3 days?

A. 2 1/2 days

✓ B. 2 days

C. 3 days

D. 3 ½ days

$$\frac{1}{t_b} = \frac{1}{6} + \frac{1}{3}$$

$$\frac{1}{t_b} = \frac{3+6}{18}$$

$$\frac{1}{t_b} = \frac{9}{18} \quad \frac{1}{t_b} = \frac{1}{2}$$

$$t_b = 2$$

Algebra Work Problem Formula

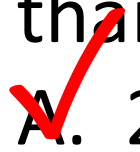
$$\frac{1}{t_b} = \frac{1}{t_1} + \frac{1}{t_2}$$

t_1 = time taken by first person

t_2 = time taken by second person

t_b = time taken if both do the work together

16. Tom and Bob have a total of 49 toys. If Bob has 5 more toys than Tom, how many toys does Tom has?

 A. 22

B. 27

C. 23

D. 26

$x = \text{no. of Tom's toys}$

$x + 5 = \text{no. of Bob's toys}$

$$x + x + 5 = 49$$

$$2x + 5 = 49$$

$$2x + 5 - 5 = 49 - 5$$

$$2x = 44$$

$$\frac{2x}{2} = \frac{44}{2}$$

$$x = 22$$

17. The fraction halfway between $\frac{3}{7}$ and $\frac{4}{7}$ is _____.

 A. $\frac{1}{2}$

B. $\frac{1}{3}$

C. $\frac{1}{4}$

D. $\frac{1}{8}$

$$\frac{\frac{3}{7} + \frac{4}{7}}{2} = \frac{\frac{7}{7}}{2} = \frac{1}{2}$$

18. In a certain school, the ratio of boys to girls is 5 is to 7. If there are 180 boys and girls in the school, how many boys are there?

A. 105

B. 90

C. 45

 D. 75

$$5x + 7x = 180$$

$$12x = 180$$

$$\frac{12x}{12} = \frac{180}{12}$$

$$x = 15$$

$$5x = 5(15)$$

$$5x = 75$$

19. A number is divided by 5. The result was subtracted by 154, and the difference is 6. What is the number?

- ☒ A. 800
- ☐ B. 160
- ☐ C. 200
- ☐ D. 400

$$\frac{x}{5} - 154 = 6$$

$$\frac{x}{5} - 154 + 154 = 6 + 154$$

$$\frac{x}{5} = 160$$

$$5\left(\frac{x}{5} = 160\right)5$$

$$x = 800$$

20. Joey gathered twice more marbles than Vic, and Tito gathered two more marbles than Vic. They gathered 26 marbles. How many marbles did Vic gather?

-  A. 6
- B. 12 x = no. of marbles gathered by Vic
- C. 8 $2x$ = no. of marbles gathered by Joey
- D. 14 $x + 2$ = no. of marbles gathered by Tito

$$x + (2x) + (x + 2) = 26$$

$$x + 2x + x + 2 = 26$$

$$4x + 2 = 26$$

$$4x + 2 - 2 = 26 - 2$$

$$4x = 24$$

$$\frac{4x}{4} = \frac{24}{4}$$

$$x = 6$$

21. In an English test, eight students obtained the following scores: 10, 15, 12, 18, 16, 24, 12, 14. What is the median score?

A. 14

☒ B. 14.5

C. 15.5

D. 15

10

12

12

14

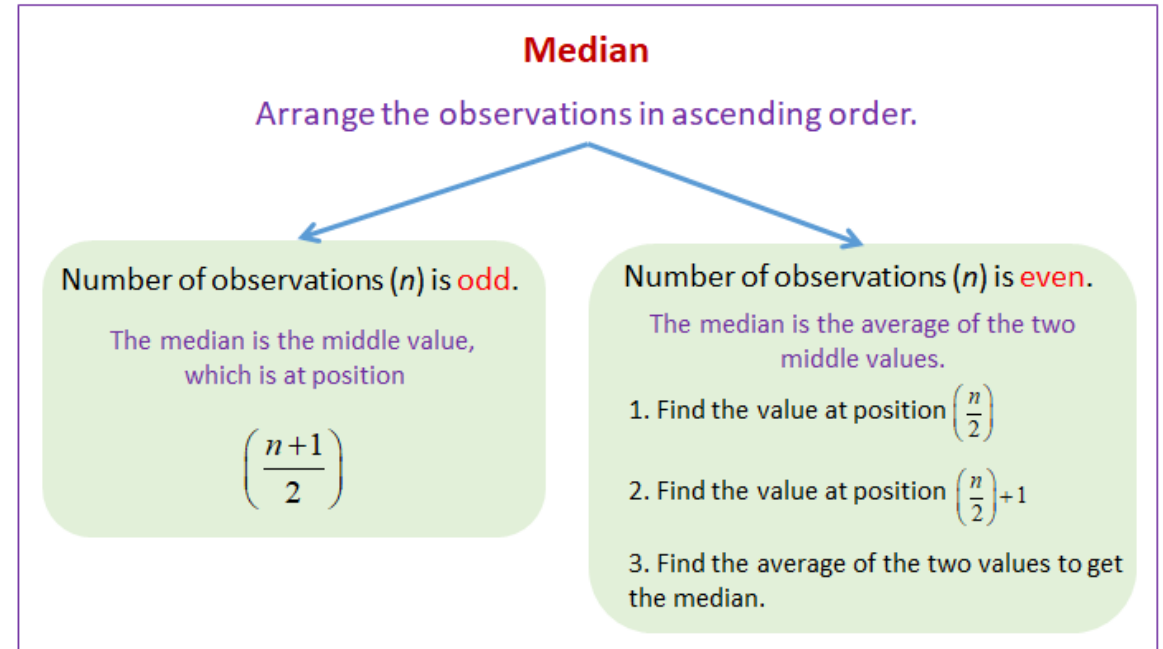
15

16

18

24

$$\frac{14 + 15}{2}$$
$$\frac{29}{2}$$
$$14.5$$



22. Find the least common multiple of 5, 2, 7

A. 35

B. 140

C. 15

 D. 70

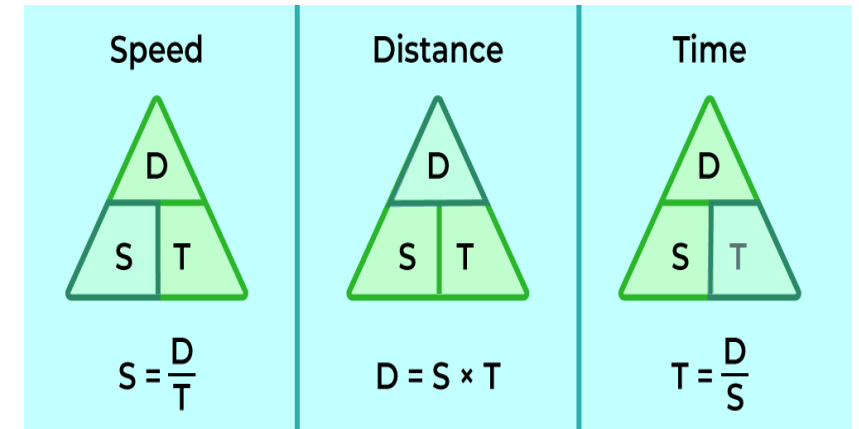
5, 2, & 7 are relatively prime

Hence, their LCM is $5 \times 2 \times 7$

$= 70$

23. A Toyota car travelling at a rate of 70 km per hour leaves the house 2 hours after a Kia car has left and overtakes it in 5 hours at what rate was the Kia car travelling?

- A. 30 kilometers per hour
- ☒ B. 50 kilometers per hour
- C. 20 kilometers per hour
- D. 40 kilometers per hour



$$S_{\text{toyota}} = 70 \text{ km per hour}$$

$$T_{\text{toyota}} = 5 \text{ hours}$$

$$D_{\text{toyota}} = S \times T$$

$$D_{\text{toyota}} = 70 \text{ km/hr} \times 5 \text{ hr}$$

$$D_{\text{toyota}} = 350 \text{ km}$$

$$S_{\text{Kia}} = ?$$

$$T_{\text{Kia}} = 5 \text{ hrs} + 2 \text{ hrs} = 7 \text{ hrs}$$

$$D_{\text{Kia}} = 350 \text{ km}$$

$$S_{\text{Kia}} = \frac{D}{T}$$

$$S_{\text{Kia}} = \frac{350 \text{ km}}{7 \text{ hrs}}$$

$$S_{\text{Kia}} = 50 \text{ km/hr}$$

24. If the average or arithmetic mean of x and -5 is 10 , then what must be the value of x ?

- A. 15
- ☒ B. 25
- C. 30
- D. 20

Mean Formula



$$\text{Mean} = \frac{\text{Sum of All Data Points}}{\text{Number of Data Points}}$$

$$10 = \frac{x + (-5)}{2}$$

$$2(10 = \frac{x + (-5)}{2})2$$

$$20 = x - 5$$

$$20 + 5 = x - 5 + 5$$

$$25 = x$$

30. What is the identity $\frac{\sin\theta}{\cos\theta}$?

- A. $\tan\theta$
- B. $\cos\theta$
- C. $\csc\theta$
- D. $\sec\theta$

$$\frac{\sin\theta}{\cos\theta} = ?$$

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos\theta = \frac{\text{adj}}{\text{hyp}}$$

$$\frac{\frac{\text{opp}}{\text{hyp}}}{\frac{\text{adj}}{\text{hyp}}} = \frac{\text{opp}}{\text{hyp}} \times \frac{\text{hyp}}{\text{adj}} = \frac{\text{opp}}{\text{adj}}$$

SIX TRIGONOMETRIC RATIOS

TRIGONOMETRIC RATIO	ABBREVIATION	DEFINITION
sine θ	$\sin \theta$	$\frac{\text{opposite}}{\text{hypotenuse}}$
cosine θ	$\cos \theta$	$\frac{\text{adjacent}}{\text{hypotenuse}}$
tangent θ	$\tan \theta$	$\frac{\text{opposite}}{\text{adjacent}}$

TRIGONOMETRIC RATIO	ABBREVIATION	DEFINITION
cosecant θ	$\csc \theta$	$\frac{\text{hypotenuse}}{\text{opposite}}$
secant θ	$\sec \theta$	$\frac{\text{hypotenuse}}{\text{adjacent}}$
cotangent θ	$\cot \theta$	$\frac{\text{adjacent}}{\text{opposite}}$

SOH- CAH- TOA

SINE-
OPPOSITE-
HYPOTENUSE

COSINE-
ADJACENT-
HYPOTENUSE

TANGENT-
OPPOSITE-
ADJACENT

CHO - SHA - CAO

COSECANT-
HYPOTENUSE-
OPPOSITE

SECANT-
HYPOTENUSE-
ADJACENT

COTANGENT-
ADJACENT-
OPPOSITE

**ALL THE HARD WORK,
ALL THE SACRIFICES,
ALL THE SLEEPLESS
NIGHTS, STRUGGLES,
DOWNFALLS, IT ALL
PAYS OFF.**